2020
MOOREBANK HIGH SCHOOL
TRIAL HSC EXAMINATION

MATHEMATICS ADVANCED

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks : 100

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7 – 23)

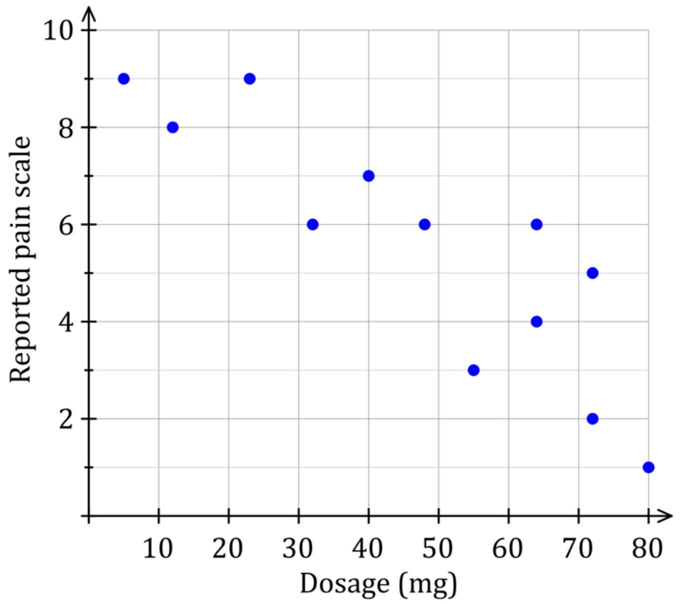
- Attempt Questions 11 – 35
- Allow about 2 hours and 45 minutes for this section

Section I (10 marks)

Attempt Questions 1–10
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

1. A scatterplot of pain (as reported by patients) compared to the dosage (in mg) of a drug is shown below.



How could you describe the correlation between the pain and the dosage?

(A) A moderate negative correlation.

(B) A moderate positive correlation.

(C) A weak positive correlation.

(D) No correlation.

2. What values of x is the curve $f(x) = 2x^3 + x^2$ concave down?

(A) $x < -\frac{1}{6}$

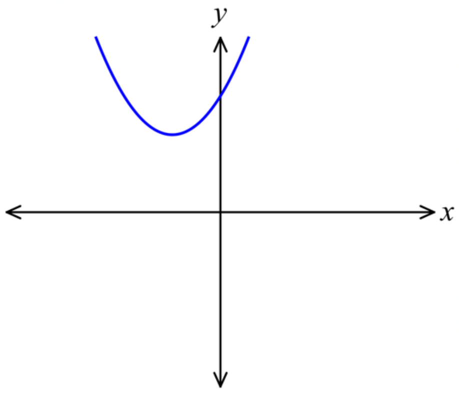
(B) $x > -\frac{1}{6}$

(C) $x < -6$

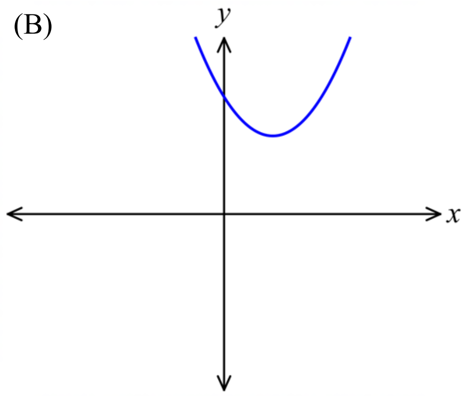
(D) $x > 6$

3. Which diagram best shows the graph of the parabola $y = 2 - (x + 1)^2$?

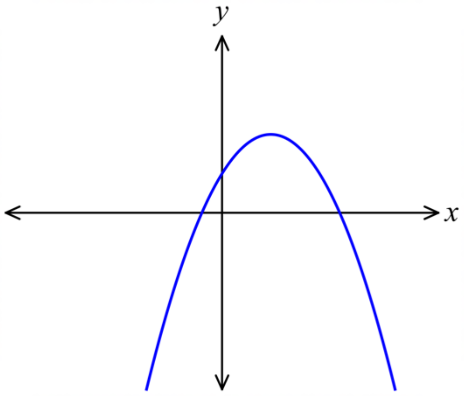
(A)



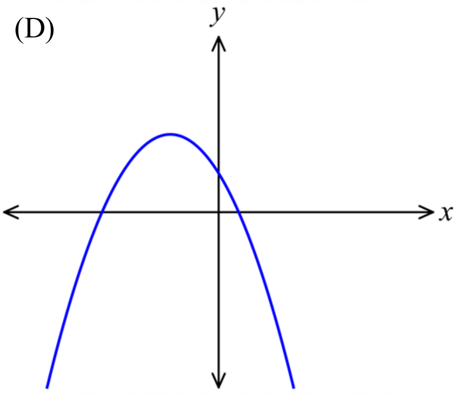
(B)



(C)



(D)



4. The weekly pay for workers at the Prosper Factory is normally distributed, with a mean of \$750 and a standard deviation of \$35.

What percentage of workers earn below \$680 a week?

(A) 0.15%

(B) 2.5%

(C) 5%

(D) 47.5%

5. The function $y = \ln x$ is transformed by first being dilated vertically by a scale factor of 3 and then translated horizontally 4 units to the left.
Find the equation of the transformed function.

(A) $y = 3 \ln x^4$

(B) $y = 4 \ln x^3$

(C) $y = 3 \ln (x - 4)$

(D) $y = 3 \ln (x + 4)$

6. The 7th term of an arithmetic sequence is 45 and the 11th term is 77.
Find the first term (a) and the common difference (d).

(A) $a = -3$ and $d = 8$

(B) $a = 3$ and $d = 8$

(C) $a = 8$ and $d = -3$

(D) $a = 8$ and $d = 3$

7. Twenty students sit a Chemistry test and the mean of their scores is 78.
Two students sit the test late and their scores are 95 and 83.
What is the new mean for the Chemistry test?

(A) 79

(B) 80

(C) 83

(D) 89

8. What is the equation of the axis of symmetry of the quadratic function $y = -k(x + b)^2 + c$?

(A) $x = 2ak$

(B) $x = b$

(C) $x = 2b$

(D) $x = -b$

9. A function is given by $f(x) = \begin{cases} \dfrac{3x^2}{125} & 0 \le x \le 5 \\ 0 & \text{elsewhere} \end{cases}$

If this function is a continuous probability distribution, what is the area under the curve?

(A) -1

(B) 0.5

(C) 1

(D) 2

10. Find the derivative of $e^{x \sin 3x}$.

(A) $e^{3x \cos 3x}$

(B) $e^{x \sin 3x}(\sin 3x + 3x \cos 3x)$

(C) $e^{x \sin 3x}$

(D) $e^{x \sin 3x}(\sin 3x - 3x \cos 3x)$

End of Section I

MOOREBANK HIGH SCHOOL

2020 TRIAL HSC EXAMINATION

Class and Teacher

MATHEMATICS ADVANCED

Student Number

Section II Answer Booklet

Student Name

90 marks
Attempt Questions 11 – 27
Allow about 2 hours and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet.
  If you use this space, clearly indicate which question you are answering.

Question 11 Differentiate the following with respect to x — Marks

a) $\ln(x^2 + 2)$ — 1

b) $\dfrac{\sin x}{x^2}$ — 2

Question 12

a) Show that the derivative of $\ln\left(\dfrac{3+x}{3-x}\right)$ is $\dfrac{6}{9-x^2}$. — 2

Question 12 continues onto the next page

b) Hence or otherwise find $\int \frac{1}{9 - x^2}\, dx$. 2

Question 13 Find the anti-derivative of $\frac{1}{1 - 2x}$ with respect to x. 2

Question 14 Let $f(x) = \frac{(x+3)(2x+1)}{\sqrt{x}}, \quad x > 0$

a) Show that $f(x)$ can be written in the form $Ax^{\frac{3}{2}} + Bx^{\frac{1}{2}} + Cx^{-\frac{1}{2}}$

Find the values of A, B, and C. 2

Question 14 continues onto the next page

b) Find $f'(x)$

1

Question 15 Find the common ratio of a geometric series with a first term of 3 and a limiting sum of 18.

2

Question 16

Find $\int 6x^2 + 2 + x^{-\frac{1}{2}} \, dx$, giving each term in its simplest form.

2

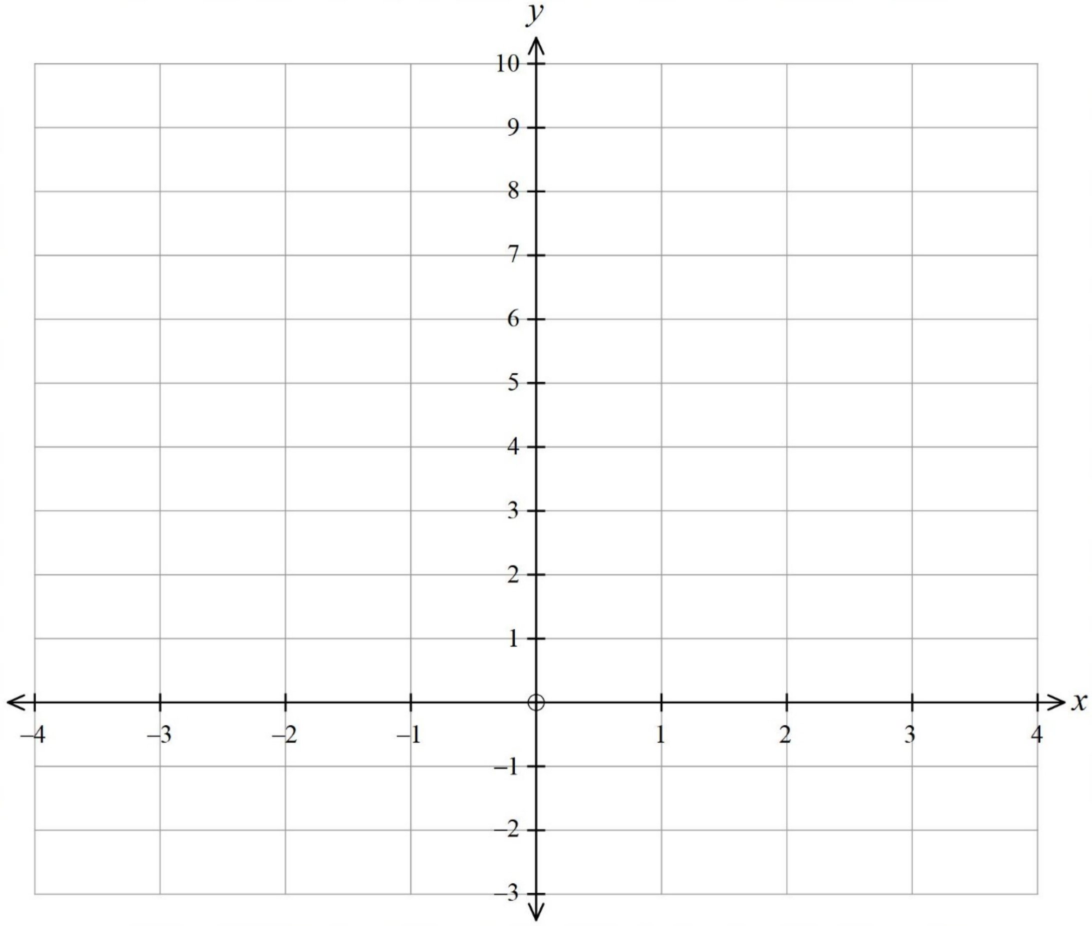
Question 17 If $\tan\theta = \frac{2}{3}$, and θ is acute, find the exact value of $\sin\theta$.

2

Question 18

a) Show that $\dfrac{4x - 3}{x} = \dfrac{-3}{x} + 4$ 1

b) Hence or otherwise, sketch the graph of $y = \dfrac{4x-3}{x}$ showing any asymptotes and the x-intercept. 2



Question 19 Fred sits his Trial exams in Modern History and Ancient History.

The marks for the Modern History class have a mean of 54 and a standard deviation of 5.6.

The marks for the Ancient History class have a mean of 76 and a standard deviation of 2.1.

a) Compare and contrast the distribution of marks for the two classes. 2

b) Fred scored 65 for Modern History and 80 for Ancient History.

Using z-score calculations, explain which subject he performed better in and why. 3

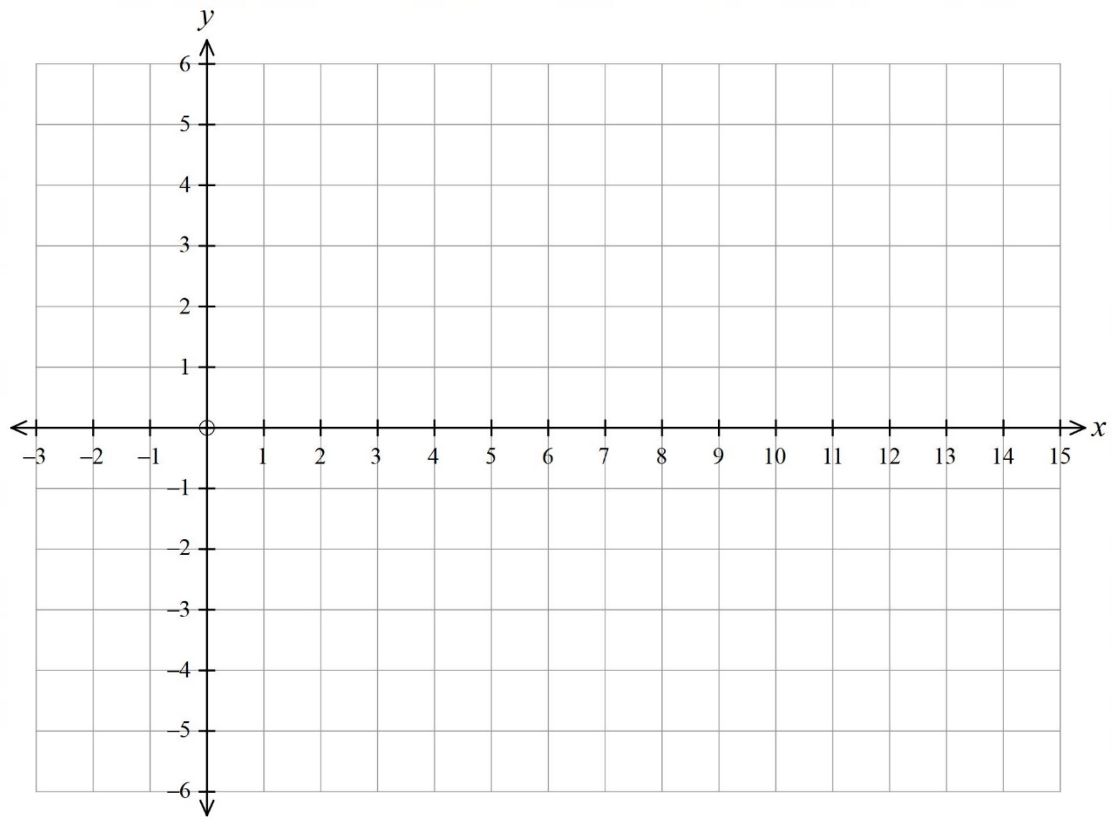
Question 20 Evaluate $\int_{1}^{e^3} \frac{5}{x}\, dx$. 2

Question 21

a) Solve the equations $y = \sqrt{x+1}$ and $y = 11 - x$ simultaneously and show that there is one point of intersection. Give its coordinates.

3

b) Sketch $y = \sqrt{x+1}$ and $y = 11 - x$ on the axes below.



2

Question 21 continues onto the next page

c) Calculate the area bounded by the curves $y = \sqrt{x+1}$ and $y = 11 - x$ and the x-axis. 2

Question 22

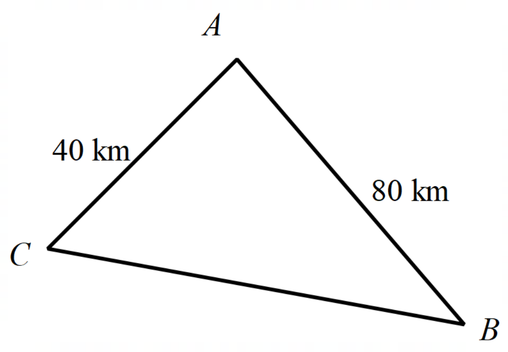
a) Given $f(x) = \sqrt{4 - x^2}$ complete this table of values, correct to 3 decimal places. 1

x	0	0.5	1	1.5	2
$f(x)$					

b) Use the Trapezoidal rule, with four sub-intervals, to estimate the value of $\int_0^2 \sqrt{4 - x^2}\, dx$. 2

Question 23 Three towns, A, B and C form a triangle.

Town A is 80 km from Town B and Town C is 40 km from Town A as shown below:



The bearing of Town B from Town A is $130°$. The bearing of Town C from Town A is $240°$.

a) Find the area of the triangle formed by the three towns, to the nearest square kilometre.

2

b) Using the cosine rule, find the distance between Town B and Town C, to the nearest kilometre.

2

Question 24 For the curve $y = x^3 - 3x^2 - 9x + 4$:

a) Find any stationary points and classify them. 3

b) Find the point of inflexion. 2

c) Sketch the curve, showing all main features. 2

Question 25 A particle is moving in a straight line with velocity $v = 3e^{t} + 6e^{-t}$ with t measured in minutes and v in ms^{-1}.

The particle begins its motion at origin.

a) What is the initial velocity? 1

b) Find an equation for x, the displacement of the particle. 2

c) Show that when $x = 10$, $3e^{2t} - 7e^{t} - 6 = 0$. 2

d) Hence, find the value of t when $x = 10$. 2

Question 26 The random variable X has this probability distribution.

X	0	1	2	3	4
$P(X = x)$	0.1	0.2	0.4	0.2	0.1

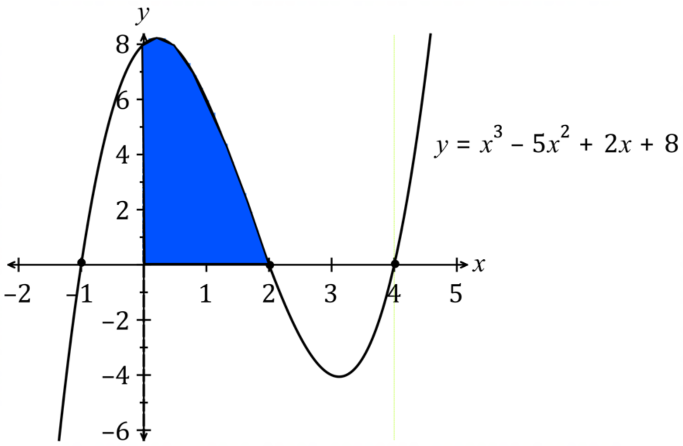
a) Find $P(1 < X \leq 3)$ 1

b) Find the variance of X. 2

Question 27

Find the exact value of $\displaystyle\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x \; dx$. 2

Question 28



Find the shaded area enclosed by the curve $y = x^3 - 5x^2 + 2x + 8$ and the coordinate axes.

2

Question 29 The probability density function for the continuous random variable X is given by:

$$f(x) = \begin{cases} |1 - x| & 0 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$$

a) Find $P\,(X \le 1.5)$

2

b) Find $P\,(1 \le X \le 2)$

2

Question 30 John started work at 18 and at the beginning of each month he invested \$400 into a superannuation fund. Interest was paid at 6% p.a. compounded monthly on the investment. John retired at 63 after having contributed to the fund for 45 years.

a) How much did John contribute to the fund over 45 years? **1**

b) How much did John's investment amount to after 45 years? **2**

c) John plans to reinvest some of the money into an account which offers 8% p.a. compound interest compounded annually. He plans to have \$300 000 at the end of the 10 year investment period. How much does John need to reinvest to achieve this amount. Answer correct to the nearest dollar. **2**

Question 31

A curve with the equation $y = f(x)$, has $\dfrac{dy}{dx} = x^3 + 2x - 7$

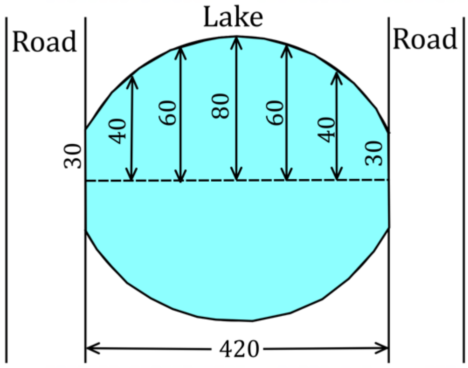
a) Find $\dfrac{d^2y}{dx^2}$ 1

b) Show that $\dfrac{d^2y}{dx^2} \geq 2$ for all values of x. 1

c) The point $P(2, 4)$ lies on the curve. Find y in terms of x. 2

d) Find an equation for the normal to the curve at P, in the form $ax + by + c = 0$, where a, b and c are integers. 2

Question 32 A symmetrical lake has two roads, 420 metres apart, forming two of its sides.



Not to scale

Equally spaced measurements of the lake, in metres, are shown on the above diagram. Use the trapezoidal rule to estimate the area of the lake.

3

Question 33 A leak from a tanker has accidently contaminated a farmer's paddock with a toxic chemical.

The chemical concentration in the soil was 6kL/ha immediately after the accident.

One year later the concentration was measured to be 2.4kL/ha.

It is known that the concentration, C, is given by:

$$C = C_0 e^{-kt}$$

Where C_0 and k are constants and t is measured in years.

Evaluate C_0 and k.

3

Question 34 The table below shows the present value interest factors for some monthly interest rates and loan periods in months.

Present value of $1				
Period	*0.0060*	*0.0065*	*0.0070*	*0.0075*
46	40.09350	39.64965	39.21263	38.78231
47	40.84841	40.38714	39.93310	39.48617
48	41.59882	41.11986	40.64856	40.18478
49	42.34475	41.84785	41.35905	40.87820

a) Find the present value, if $3200 is contributed per month for 46 months at 0.75% per month. Answer to the nearest cent. 1

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b) Annabelle borrows $27 000 for a car. She arranges to repay the loan with monthly repayments over 4 years. She is charged 7.8% per annum interest. Find Annabelle's monthly repayment. Answer to the nearest cent. 2

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Question 35 The average life cycle of an insect is one month. A viable nest of this insect has between 100 000 to 500 000 insects. The population P of a nest of this insect grows exponentially so that:

$$\frac{dP}{dt} = 1200e^{0\cdot 3t}$$

A nest of these insects had a population of 5000 after one month.
Determine how long it will take the nest to reach the viable stage (i.e. when the population has reached 100 000). Answer correctly to the nearest month.

3

End of paper

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

2020 Trial HSC Examination

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a+b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

and $\alpha\beta\gamma = -\frac{d}{a}$

Relations

$$(x-h)^2 + (y-k)^2 = r^2$$

Financial Mathematics

$$A = P(1+r)^n$$

Sequences and series

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}, \ r \neq 1$$

$$S = \frac{a}{1-r}, \ |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

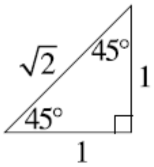
$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab\sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

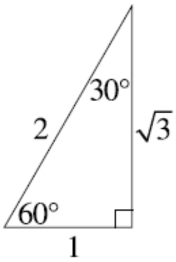


$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \ \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \ \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \ \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A\cos B + \cos A\sin B$$

$$\cos(A + B) = \cos A\cos B - \sin A\sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$

If $t = \tan\frac{A}{2}$ then $\sin A = \frac{2t}{1+t^2}$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A\cos B = \frac{1}{2}\left[\cos(A - B) + \cos(A + B)\right]$$

$$\sin A\sin B = \frac{1}{2}\left[\cos(A - B) - \cos(A + B)\right]$$

$$\sin A\cos B = \frac{1}{2}\left[\sin(A + B) + \sin(A - B)\right]$$

$$\cos A\sin B = \frac{1}{2}\left[\sin(A + B) - \sin(A - B)\right]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

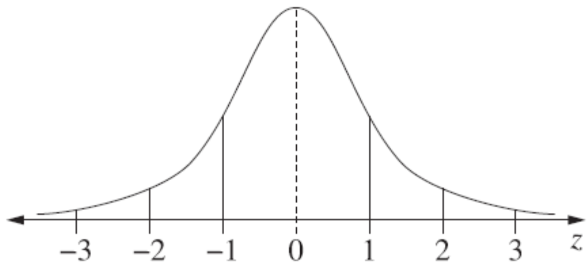
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between –1 and 1
- approximately 95% of scores have z-scores between –2 and 2
- approximately 99.7% of scores have z-scores between –3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E\left[(X - \mu)^2\right] = E\left(X^2\right) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \ P(B) \neq 0$$

Continuous random variables

$$P(X \leq x) = \int_a^x f(x)\,dx$$

$$P(a < X < b) = \int_a^b f(x)\,dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r(1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow \ P(X = x)$$

$$= \binom{n}{x}p^x(1 - p)^{n-x}, \ x = 0, 1, \ldots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function	Derivative
$y = f(x)^n$	$\dfrac{dy}{dx} = nf'(x)[f(x)]^{n-1}$
$y = uv$	$\dfrac{dy}{dx} = u\dfrac{dv}{dx} + v\dfrac{du}{dx}$
$y = g(u)$ where $u = f(x)$	$\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$
$y = \dfrac{u}{v}$	$\dfrac{dy}{dx} = \dfrac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}$
$y = \sin f(x)$	$\dfrac{dy}{dx} = f'(x)\cos f(x)$
$y = \cos f(x)$	$\dfrac{dy}{dx} = -f'(x)\sin f(x)$
$y = \tan f(x)$	$\dfrac{dy}{dx} = f'(x)\sec^2 f(x)$
$y = e^{f(x)}$	$\dfrac{dy}{dx} = f'(x)e^{f(x)}$
$y = \ln f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{f(x)}$
$y = a^{f(x)}$	$\dfrac{dy}{dx} = (\ln a)f'(x)a^{f(x)}$
$y = \log_a f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{(\ln a)f(x)}$
$y = \sin^{-1} f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{\sqrt{1-[f(x)]^2}}$
$y = \cos^{-1} f(x)$	$\dfrac{dy}{dx} = -\dfrac{f'(x)}{\sqrt{1-[f(x)]^2}}$
$y = \tan^{-1} f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{1+[f(x)]^2}$

Integral Calculus

$$\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x)\sin f(x)dx = -\cos f(x) + c$$

$$\int f'(x)\cos f(x)dx = \sin f(x) + c$$

$$\int f'(x)\sec^2 f(x)dx = \tan f(x) + c$$

$$\int f'(x)e^{f(x)}dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + c$$

$$\int f'(x)a^{f(x)}dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2-[f(x)]^2}}dx = \sin^{-1}\frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2+[f(x)]^2}dx = \frac{1}{a}\tan^{-1}\frac{f(x)}{a} + c$$

$$\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$$

$$\int_a^b f(x)dx$$

$$\approx \frac{b-a}{2n}\left\{f(a) + f(b) + 2\left[f(x_1) + \cdots + f(x_{n-1})\right]\right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^{n}P_{r} = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$\left|\underset{\sim}{u}\right| = \left|x\underset{\sim}{i} + y\underset{\sim}{j}\right| = \sqrt{x^2 + y^2}$$

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = \left|\underset{\sim}{u}\right|\left|\underset{\sim}{v}\right|\cos\theta = x_1x_2 + y_1y_2,$$

where $\underset{\sim}{u} = x_1\underset{\sim}{i} + y_1\underset{\sim}{j}$

and $\underset{\sim}{v} = x_2\underset{\sim}{i} + y_2\underset{\sim}{j}$

$$\underset{\sim}{r} = \underset{\sim}{a} + \lambda\underset{\sim}{b}$$

Complex Numbers

$$z = a + ib = r(\cos\theta + i\sin\theta)$$
$$= re^{i\theta}$$

$$\left[r(\cos\theta + i\sin\theta)\right]^n = r^n(\cos n\theta + i\sin n\theta)$$
$$= r^n e^{in\theta}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v\frac{dv}{dx} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$$

$$x = a\cos(nt + \alpha) + c$$

$$x = a\sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

MOOREBANK HIGH SCHOOL

2020 Trial Higher School Certificate Examination

Mathematics Advanced

Name ________________________________ Teacher ________________________

Section I – Multiple Choice Answer Sheet

Allow about 15 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: 2 + 4 = (A) 2 (B) 6 (C) 8 (D) 9

A ○ B ● C ○ D ○

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ● B ⊗ C ○ D ○

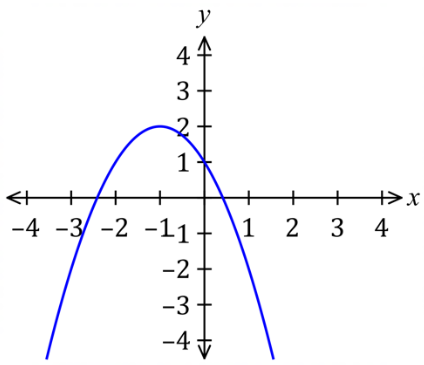
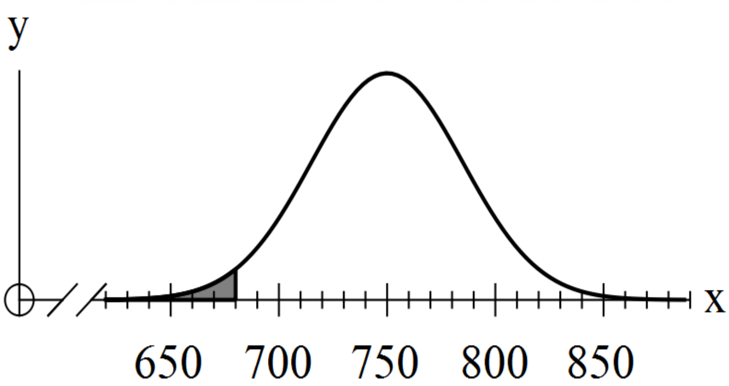
If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

correct

A ⊗ B ⊗ C ○ D ○

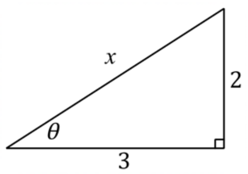
1. A ○ B ○ C ○ D ○
2. A ○ B ○ C ○ D ○
3. A ○ B ○ C ○ D ○
4. A ○ B ○ C ○ D ○
5. A ○ B ○ C ○ D ○
6. A ○ B ○ C ○ D ○
7. A ○ B ○ C ○ D ○
8. A ○ B ○ C ○ D ○
9. A ○ B ○ C ○ D ○
10. A ○ B ○ C ○ D ○

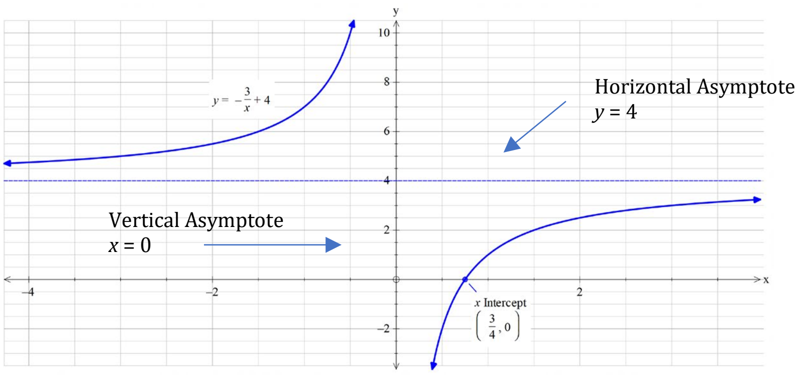
2020 Year 12 Mathematics Advanced Trials Solutions

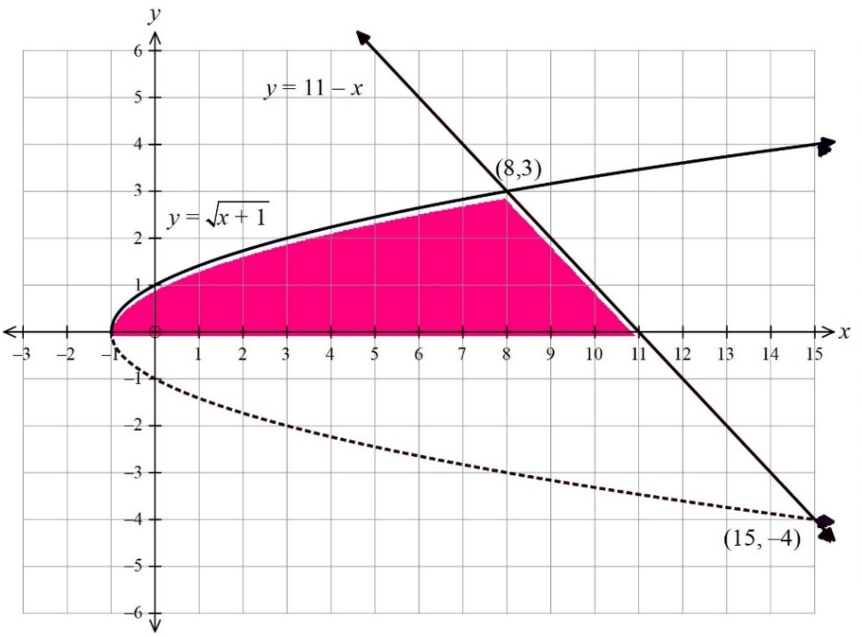
Section I		Criteria
1.	A moderate negative correlation.	1 Mark: A
2.	$f(x) = 2x^3 + x^2$ $f'(x) = 6x^2 + 2x$ $f''(x) = 12x + 2$ Concave down $f''(x) < 0$ $f''(x) < 0$ $12x + 2 < 0$ $x < -\frac{1}{6}$	1 Mark: A
3.		1 Mark: D
4.	< $ Th  v $680.	1 Mark: B
5.	$y = \ln x$ $y = 3\ln x$ vertical dilation of 3 $y = 3\ln(x + 4)$ horizontal translation to left 4 units	1 Mark: D

6.	$T_n = a + (n-1)d$ $T_7 = 45\ T_n = a + 6d$ (1) $T_{11} = 77\ T_n = a + 10d$ (2) (2) − (1) $4d = 32$ $d = 8$ (3) sub into (1) $45 = a + 6 \times 8$ $a = -3$ $a = -3$ and $d = 8$	1 Mark: A
7.	$78 = \dfrac{\Sigma x}{20}$ $\Sigma x = 78 \times 20$ $= 1560$ $\bar{x}_{\text{new}} = \dfrac{1560 + 95 + 83}{22}$ $\bar{x}_{\text{new}} = 79$	1 Mark: A
8.	Since $(x + b)$ parabola is translated b units to the left, so axis is $x = -b$ OR Axis of symmetry $x = -\dfrac{b}{2a}$ $y = -k(x+b)^2 + c$ $= -k(x^2 + 2xb + b^2) + c$ $= -kx^2 - 2kbx - kb^2 + c$ $x = \dfrac{--2kb}{-2k}$ $= -b$	1 Mark: D
9.	Area under the curve must equal 1. $A = \displaystyle\int_0^5 \frac{3x^2}{125}\,dx$ $= \dfrac{3}{125}\displaystyle\int_0^5 x^2\,dx$ $= \dfrac{3}{125}\left[\dfrac{x^3}{3}\right]_0^5$ $= \dfrac{3}{125} \times \dfrac{125}{3} - 0$ $= 1$	1 Mark: C

10.	$\frac{d}{dx}(e^{x\sin 3x}) = (e^{x\sin 3x}) \times \frac{d}{dx}(x\sin 3x)$ Find $\frac{d}{dx}(x\sin 3x)$ $u = x \quad v = \sin 3x$ $u' = 1 \quad v' = 3\cos 3x$ $vu' + uv' = \sin 3x + 3x\cos 3x$ $\frac{d}{dx}(e^{x\sin 3x}) = (e^{x\sin 3x})(\sin 3x + 3x\cos 3x)$	1 Mark: B
Section II		Criteria
11. (a)	$\frac{d}{dx}\ln(x^2+2) = \frac{2x}{x^2+2}$	1 Mark: Correct answer.
(b)	$\frac{d}{dx}\left(\frac{\text{sin}x}{x^2}\right) = \frac{x^2\text{cos}x - \text{sin}x \times 2x}{(x^2)^2}$ $= \frac{x\text{cos}x - 2\text{sin}x}{x^3}$	2 Marks: Correct answer. 1 Mark: Uses the quotient rule correctly.
12. (a)	$y = \left[\ln\left(\frac{3+x}{3-x}\right)\right]$ $y = \ln(x+3) - \ln(3-x)$ $y' = \frac{1}{3+x} - \frac{-1}{3-x}$ $= \frac{3-x+3+x}{9-x^2}$ $= \frac{6}{9-x^2}$	2 Marks: for correct solution 1 Mark: for correct use of log laws or correct steps in differentiation or equivalent merit
(b)	$\int \frac{1}{9-x^2}\,dx$ $= \frac{1}{6}\int \frac{6}{9-x^2}\,dx$ $= \frac{1}{6}\ln\left[\frac{3+x}{3-x}\right] + c$	2 Marks: for correct solution 1 Mark: for taking out the common factor to get part (a) answer or equal merit
13.	$\int \frac{1}{1-2x}\,dx = -\frac{1}{2}\ln\|1-2x\|$	2 Marks: Correct answer. 1 Mark: Finds the integral as a log function.
14. (a)	$f(x) = \frac{(x+3)(2x+1)}{\sqrt{x}}$ $= \frac{2x^2+7x+3}{\sqrt{x}}$ $= 2x^{\frac{3}{2}} + 7x^{\frac{1}{2}} + 3x^{-\frac{1}{2}}$ $\therefore A = 2, B = 7$ and $C = 3$	2 Marks: Correct answer. 1 Mark: Finds *A* or *B* or *C*.

(b)	$f(x) = 2x^{\frac{3}{2}} + 9x^{\frac{1}{2}} + 3x^{-\frac{1}{2}}$ $f'(x) = 3x^{\frac{1}{2}} + \frac{9}{2}x^{-\frac{1}{2}} - \frac{3}{2}x^{-\frac{3}{2}}$	1 Mark: Correct answer.
15.	$S_n = \frac{a}{1-r}$ $18 = \frac{3}{1-r}$ $18 - 18r = 3$ $18r = -15$ $r = -\frac{15}{18} = -\frac{5}{6}$ $\therefore$ The common ratio is $-\frac{5}{6}$	2 Marks: Correct answer. 1 Mark: Uses the limiting sum with at least one correct value.
16.	$\int 6x^2 + 2 + x^{-\frac{1}{2}}\,dx = \frac{6x^3}{3} + 2x + \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C$ $= 2x^3 + 2x + 2x^{\frac{1}{2}} + C$	2 Marks: Correct answer. 1 Mark: Integrates one term correctly.
17.	$x^2 = 2^2 + 3^2$ $x = \sqrt{13}$ $\sin\theta = \frac{2}{\sqrt{13}}$ 	2 Marks: Correct answer. 1 Mark: Finds the value of *x* or makes some progress.
18. (a)	$\frac{4x-3}{x} = \frac{4x}{x} - \frac{3}{x}$ $= 4 - \frac{3}{x}$ $= -\frac{3}{x} + 4$	1 Mark: for correct solution

(b)		2 Marks: for diagram with correct shape, asymptotes and *x*-intercept 1 Mark: if shape, asymptotes or *x*-intercept are incorrect
19. (a)	The Ancient History class has a higher mean and a smaller standard deviation, meaning the scores are grouped more closely around the mean which indicates more consistent marks. While the Modern History has a lower mean and a larger standard deviation, meaning the scores are more spread out around the lower mean, so are much less consistent marks.	2 Marks: if correctly commented on both mean and standard deviation 1 Mark: if only comment on either mean or standard deviation
(b)	$z = \frac{x - \mu}{\sigma}$ MH $x = 65 \quad \sigma = 5.6 \quad \mu = 54$ $z_{MH} = \frac{65 - 54}{5.6} = 1.96428571$ AH $x = 80 \quad \sigma = 2.1 \quad \mu = 76$ $z_{AH} = \frac{80 - 76}{2.1} = 1.9047619$ Fred performed better in MH as he had a higher positive z -score meaning he performed further above the mean, compared to AH, even though he had a lower raw mark for MH.	3 Marks: for two correct z–score calculations and analysis 2 Marks: for correct z scores but incorrect analysis or equivalent merit 1 Mark: for a correct z – score calculation or equivalent merit
20.	$\int_1^{e^3} \frac{5}{x}\,dx = 5\int_1^{e^3} \frac{1}{x}\,dx$ $= 5\left[\ln x\right]_1^{e^3}$ $= 5(\ln e^3 - \ln 1)$ $= 5(3\ln e - 0)$ $= 15$	2 Marks: for correct solution 1 Mark: for correct integration with an error in evaluation or equivalent merit

21. (a)	$$\begin{aligned} \sqrt{x+1} &= 11-x \quad \text{squaring both sides} \\ x+1 &= (11-x)^2 \\ x+1 &= 121 - 22x + x^2 \\ x^2 - 23x + 120 &= 0 \\ x^2 - 8x - 15x + 120 &= 0 \\ (x-8)(x-15) &= 0 \\ x &= 8 \text{ or } 15 \\ \text{When } x = 8 \quad y &= 3 \\ \text{When } x = 15 \quad y &= -4 \end{aligned}$$ *Other methods of solution are possible.* *Since the radical sign implies positive square root,* $y = \sqrt{x+1}$ *only exists above the x axis, so the point (15 , -4) is not part of the solution The only point of intersection is (8, 3).*	3 Marks: for correct solution including eliminating the incorrect solution. 2 Marks: for giving both solutions with no explanation or equivalent merit 1 Mark: for some relevant working on solving the equations
(b)		2 Marks: for two correct graphs (shading not required, nor is the dotted curve) 1 Mark: for only one graph correct or equivalent merit

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<td>(c)</td>
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$$A = \int_{-1}^{8} \sqrt{x+1}\ dx + \int_{8}^{11} 11 - x\ dx$$
$$= \int_{-1}^{8} (x+1)^{\frac{1}{2}}\ dx + \int_{8}^{11} 11 - x\ dx$$
$$= \left[\frac{2(x+1)^{\frac{3}{2}}}{3}\right]_{-1}^{8} + \left[11x - \frac{x^2}{2}\right]_{8}^{11}$$
$$= \left[\frac{2(8+1)^{\frac{3}{2}}}{3} - \frac{2(-1+1)^{\frac{3}{2}}}{3}\right] + \left[\left(121 - \frac{121}{2}\right) - \left(88 - \frac{64}{2}\right)\right]$$
$$= 18 + 4.5$$
$$= 22.5\ units^2$$

Students could also find simple
area of a triangle for the 2nd section of the integral

$$A = \frac{1}{2} \times 3 \times 3 = 4.5 units^2$$

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<td>2 Marks: for finding the required area by any valid method.

1 /mark: for finding the area under one of the curves or equivalent merit</td>
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<tr>
<td>22.
(a)</td>
<td>

x	0	0.5	1	1.5	2
$f(x)$	2	1.936	1.732	1.323	0

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<td>1 Mark: for ALL correct answers</td>
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<td>(b)</td>
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$$\int_a^b f(x)dx \approx \frac{b-a}{2n}[y_0 + y_n + 2 \times (y_1 + y_2 + \cdots y_{n-1})]$$
$$\approx \frac{2}{8}[(2+0) + 2(1.936 + 1.732 + 1.323)]$$
$$\approx 2.9955$$

</td>
<td>2 Marks: for correct estimate using formula or 4 trapezia

1 Mark: for some progress by any method</td>
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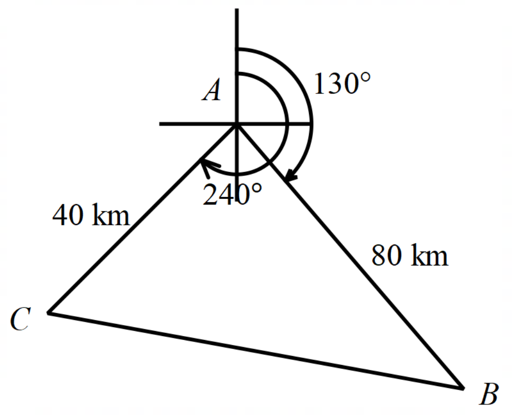
<td>23.
(a)</td>
<td>



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<td>2 Marks: for finding correct area

1 Mark: for finding the angle CAB from the bearings or equivalent merit</td>
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$\angle CAB = 240° - 130° = 110°$

$$A = \frac{1}{2} \times 40 \times 80 \times \sin 110°$$
$$= 1503.508193$$
$$= 1504 \text{ km}^2$$

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<td>(b)</td>
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$$a^2 = 40^2 + 80^2 - 2 \times 40 \times 80 \cos 110°$$
$$= 10188.92892$$
$$a = 100.9402245$$
$$a = 101 \text{ km}$$

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<td>2 Marks: for correct answer

1 Mark for the correct substitution or equivalent merit</td>
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<td>24.
(a)</td>
<td>

$$y = x^3 - 3x^2 - 9x + 4$$
$$y' = 3x^2 - 6x - 9$$
$$= 3(x^2 - 2x - 3)$$
$$= 3(x-3)(x+1)$$
$$y'' = 6x - 6$$
$$= 6(x-1)$$

Stat pts when $y' = 0$

$3(x-3)(x+1) = 0$

$x = 3$ or $x = -1$

when $x = 3$ $y = -23$

$y'' = 12$ $> 0 \therefore$ minimum

when $x = -1$ $y = 9$

$y'' -12$ $< 0 \therefore$ maximum

</td>
<td>3 Marks: for correct points and (checked) nature

2 Marks: for finding y’ and y” and turning points but not classifying or equivalent merit

1 Mark: for finding y’ and y” or equivalent merit</td>
</tr>
<tr>
<td>(b)</td>
<td>

Inflexion when $y'' = 0$

$6(x-1) = 0$

$x = 1$

when $x = 1$ $y = -7$

check concavity change

x	0	1	2
y”	-6	0	6

Concavity changes so (1, -7) is an inflexion.

</td>
<td>2 Marks: for correct point and working out

1 Mark: for correct point</td>
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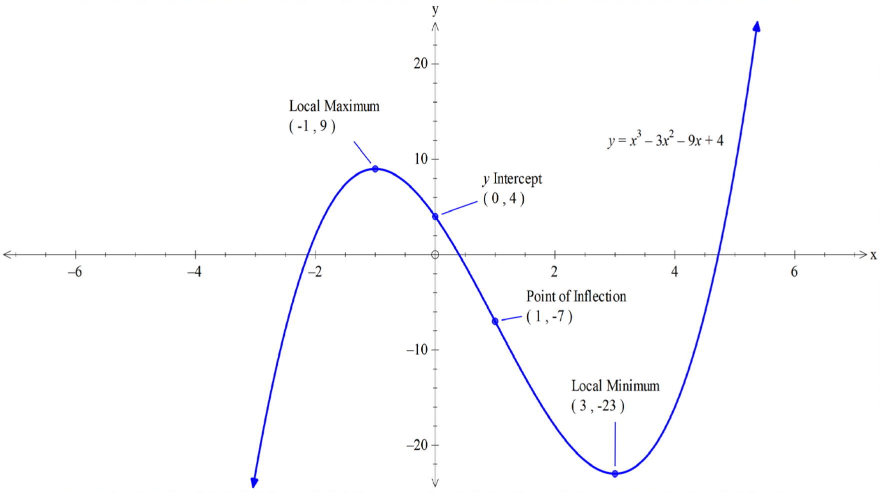
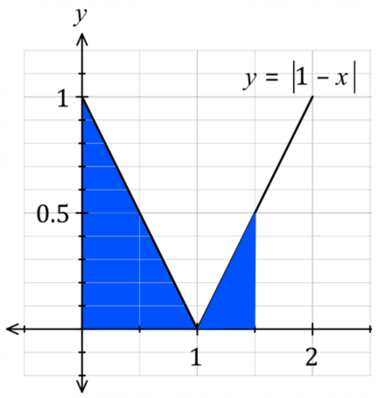
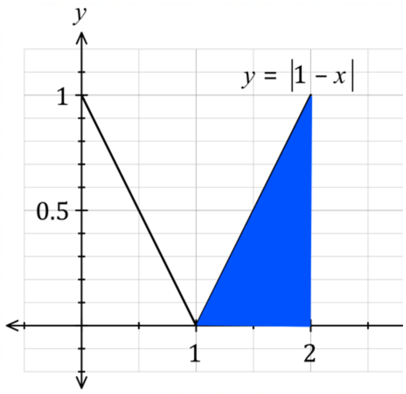
(c)		2 Marks: for correct shape of the curve and the 4 points shown. 1 Mark: for an incorrect graph with some points correct
25. (a)	$v = 3e^t + 6e^{-t}$ when $t = 0$ $v = 3e^0 + 6e^0$ $= 9ms^{-1}$	1 Mark: for correct answer
(b)	$x = \int v dt$ $= 3e^t + 6e^{-t}$ dt $= 3e^t - 6e^{-t} + c$ When $t = 0 \quad x = 0$ $0 = 3 - 6 + c$ $c = 3$ $\therefore \quad x = 3e^t - 6e^{-t} + 3$	2 Marks: for giving the correct equation 1 Mark: for correct integration but not finding C or equivalent merit
(c)	When $x = 10$ $10 = 3e^t - 6e^{-t} + 3$ $7 = 3e^t - 6e^{-t}$ $7 = 3e^t - \frac{6}{e^t}$ multiplying by e^t $7e^t = 3(e^t)^2 - 6$ $3e^{2t} - 7e^t - 6 = 0$	2 Marks: for correct substitution and manipulation to show required equation 1 Mark: correct substitution and some manipulation

(d)	Let $u = e^t$ $3u^2 - 7u - 6 = 0$ $3u^2 - 9u + 2u - 6 = 0$ $3u(u-3) + 2(u-3) = 0$ $(3u+2)(u-3) = 0$ $\therefore \quad u = -\frac{2}{3}$ or 3 $e^t \neq -\frac{2}{3}$ does not exist $e^t = 3$ taking ln of both sides $t = \ln 3$ minutes	2 Marks: for substitution and solving a quadratic and finding 2 solutions 1 Mark: for substitution and attempt at solving the quadratic
26. (a)	$P(1 < X \leq 3) = 0.4 + 0.2$ $= 0.6$	1 Mark: Correct answer.
(b)	$\mu = 0 \times 0.1 + 1 \times 0.2 + 2 \times 0.4 + 3 \times 0.2 + 4 \times 0.1$ $= 2$ $Var(X) = E(X^2) - \mu^2$ $= 0^2 \times 0.1 + 1^2 \times 0.2 + 2^2 \times 0.4 + 3^2 \times 0.2 + 4^2 \times 0.1 - 2^2$ $= 1.2$ $\therefore$Variance is 1.2	2 Marks: Correct answer. 1 Mark: Shows some understanding
27.	$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x \, dx = [\sin x]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$ $= \sin\left(\frac{\pi}{2}\right) - \sin\left(\frac{\pi}{4}\right)$ $= 1 - \frac{1}{\sqrt{2}}$ $= \frac{\sqrt{2} - 1}{\sqrt{2}}$ $= \frac{2 - \sqrt{2}}{2}$	2 Marks: Correct answer. 1 Mark: Correctly integrates.
28.	$A = \int_0^2 x^3 - 5x^2 + 2x + 8dx$ $= \left[\frac{1}{4}x^4 - \frac{5}{3}x^3 + x^2 + 8x\right]_0^2$ $= \left(\frac{1}{4}(2)^4 - \frac{5}{3}(2)^3 + 2^2 + 8 \times 2\right) - 0$ $= \frac{32}{3} = 10\frac{2}{3}$	2 Marks: Correct answer. 1 Mark: Integrates one term correctly.

29. (a)	$P(X \leq 1.5)$ $= \frac{1}{2} \times 1 \times 1 + \frac{1}{2} \times 0.5 \times 0.5$ $= 0.625$  	2 Marks: Correct answer. 1 Mark: Sketches the function or shows some understanding.
(b)	$P(1 \leq X \leq 2)$ $= \frac{1}{2} \times 1 \times 1$ $= 0.5$  	2 Marks: Correct answer. 1 Mark: Sketches the function or shows some understanding.
30. (a)	Investment $= 400 \times 12 \times 45$ $= \$216\ 000$ $\therefore$ John contributed \$216 00 over the 45 years.	1 Mark: Correct answer.
(b)	$r = \frac{0.06}{12} = 0.005\%$ per month After 1 month $A_1 = (400 \times 1.005^1)$ After 2 months $A_2 = (400 \times 1.005^2) + (400 \times 1.005^1)$ After 45 years (540 months) $A_{540} = 400 \times 1.005^{540} + (400 \times 1.005^{539}) \ldots + (400 \times 1.005^1)$ Use the formula for the sum of a GP $A_{540} = 400 \times \frac{1.005(1.005^{540} - 1)}{1.005 - 1}$ $= \$1\ 107\ 909.03$ $\therefore$ John's investment is worth \$1 107 909.03 after 45 years.	2 Marks: Correct answer. 1 Mark: Makes some progress towards the solution.

(c)	$FV = PV(1+r)^n$ $300\ 000 = PV(1+0.08)^{10}$ $PV = 138\ 958.0464\ldots$ $\approx \$138\ 958$ $\therefore$ John needs to reinvest \$138 958.	2 Marks: Correct answer. 1 Mark: Uses the *FV* formula with one correct value.
31. (a)	$\frac{dy}{dx} = x^3 + 2x - 7$ $\frac{d^2y}{dx^2} = 3x^2 + 2$	1 Mark: Correct answer.
(b)	Since $x^2 \geq 0$ (for all values of x) then $3x^2 \geq 0$ $\therefore \frac{d^2y}{dx^2} \geq 2$	1 Mark: Correct answer.
(c)	Finding the anti-derivative. $y = \frac{x^4}{4} + x^2 - 7x + C$ $P(2, 4)$ is on the curve and satisfies the equation. $4 = \frac{2^4}{4} + 2^2 - 14 + C$ $C = 10$ $\therefore y = \frac{x^4}{4} + x^2 - 7x + 10$	2 Marks: Correct answer. 1 Mark: Finds the anti-derivative.
(d)	Gradient of the tangent at $P(2, 4)$ $m = \frac{dy}{dx} = 2^3 + 2 \times 2 - 7 = 5$ Gradient of the normal at $P(2, 4)$ $m_1 m_2 = -1$ $m = -\frac{1}{5}$ Equation of the normal at $P(2, 4)$ $y - y_1 = m(x - x_1)$ $y - 4 = -\frac{1}{5}(x - 2)$ $\therefore x + 5y - 22 = 0$	2 Marks: Correct answer. 1 Mark: Finds the gradient of the tangent at $P(2, 4)$
32.	Width of the strip $h = \frac{b-a}{n} = \frac{420-0}{6} = 70$ Top half of the lake. $A \approx \frac{h}{2}[y_0 + y_6 + 2(y_1 + y_2 + y_3 + y_4 + y_5)$ $\approx \frac{70}{2}[30 + 30 + 2(40 + 60 + 80 + 60 + 40)]$ $\approx 21\ 700$ Area of the entire lake $A \approx 2 \times 21\ 700 \approx 43\ 400$ m $\therefore$ Area of the lake is about 43 400 metres.	3 Marks: Correct answer. 2 Marks: Makes significant progress. 1 Mark: Uses the trapezoidal rule with at least one correct value.

33.	$C = C_0 e^{-kt}$ At $t = 0$ $C = 6$ $6 = C_0 e^0$ $C_0 =$ 6kL/ha So $C = 6e^{-kt}$ At $t = 1$ $C = 2.4$ $2.4 = 6e^{-k(1)}$ $2.4 = 6e^{-k}$ $0.4 = e^{-k}$ (taking ln of both sides) $k = -\ln(0.4)$ $k = 0.9162097319$ $k = 0.916$	3 Marks: for finding C_0 and writing a new equation for *C* and finding *k* 2 Marks: for finding C_0 writing a new equation for *C* and attempting to solve 1 Mark: for some relevant calculations and exponential manipulation
34. (a)	Intersection value is 38.78231 (0.0075 and 46 months) $PV = 38.78231 \times 3200$ $= 124\,103.392$ $\approx \$124\,103.39$	1 Mark: Correct answer.
(b)	$r = \frac{0.078}{12} = 0.0065 \quad n = 4 \times 12 = 48$ Intersection value is 41.11986 Let the monthly repayment be x. $PV = 41.11986 \times x$ $27\,000 = 41.11986 \times x$ $x = \frac{27\,000}{41.11986}$ $= 656.6170 \ldots \approx \$656.62$ $\therefore$ Annabelle's monthly repayment is \$656.62.	2 Marks: Correct answer. 1 Mark: Finds the intersection value or shows some understanding.
35.	$\frac{dP}{dt} = 1200e^{0 \cdot 3t}$ $P = \frac{1200}{0.3} e^{0 \cdot 3t} + C$ $P = 4000e^{0 \cdot 3t} + C$ Given $P = 5000$ when $t = 1$ $5000 = 4000e^{0 \cdot 3t} + C$ $C = -399.4352 \ldots \approx -399.4$ To find t when $P = 100\,000$	3 Marks: Correct answer. 2 Marks: Makes significant progress. 1 Mark: Finds the anti-derivative of the differential equation.

	≈ 11 $\therefore$ The nest reaches a viable stage after 11 months.	